ABSTRACT

Smart Restaurant Hygiene Monitoring System Using AI and Machine Learning

In this project, I aim to develop a smart system that monitors the hygiene standards of restaurants using Artificial Intelligence (AI) and Machine Learning (ML) techniques. As cleanliness and customer satisfaction are crucial in the food industry, I propose an AI-based solution to address the limitations of manual inspection, which is often time-consuming and inconsistent.

The system collects various types of data from the restaurant environment, such as kitchen images, plate images (to detect food waste), customer facial expressions, noise levels, temperature, staff presence, and crowd conditions. I use Convolutional Neural Networks (CNN) to analyze images and classify kitchen cleanliness and customer emotions. For sound analysis, I apply audio classification techniques to detect crowd or noise levels. Finally, I combine all these features and use machine learning algorithms such as Random Forest and Decision Tree to predict an overall hygiene score (ranging from 1 to 5).

This AI-driven system helps automate the evaluation of hygiene conditions in restaurants and reduces the need for continuous manual inspection. It can be used by restaurant managers to improve service quality, and more importantly, it has the potential to assist government health departments in monitoring hygiene compliance across multiple restaurants remotely. In the future, I plan to enhance the system by integrating IoT devices for real-time monitoring and automated alerts.

Reference Paper: **Application of Deep Learning in Food: A Review**

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